



MiDAS1.0 IP basic specification

V1.1

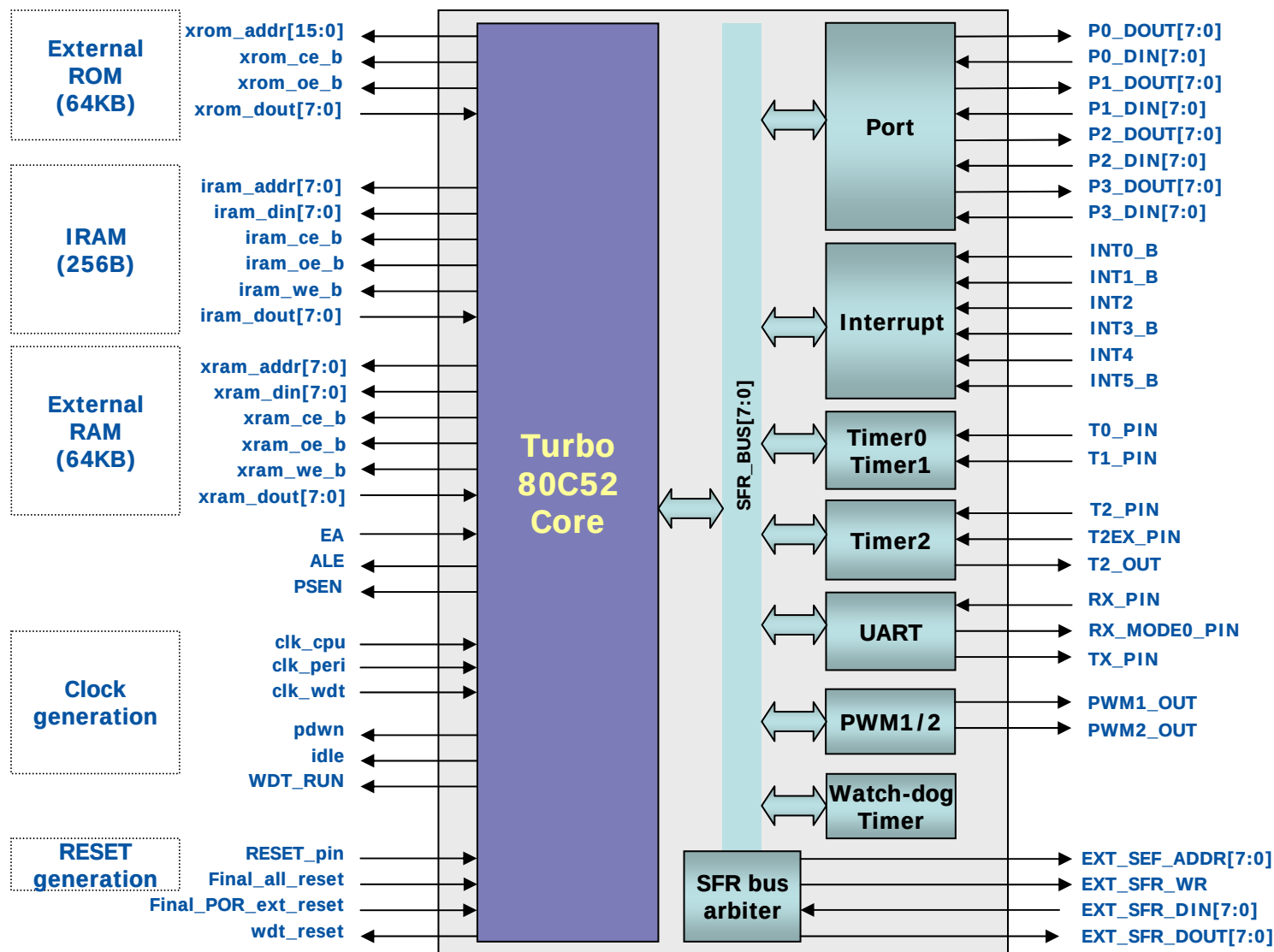
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- ◆ CPU
 - ✓ 8-bit turbo 80C52 architecture
 - ✓ 4 cycles/1 machine cycle
 - ✓ Instruction level compatible with Intel 80C52
- ◆ Max 64Kbytes program memory interface
- ◆ Max 64Kbytes data memory interface
- ◆ Fully programmable 32 IO pins
 - ✓ Optional Input/Output control
- ◆ Three 16-bit Timer/Counters
- ◆ 2 channel 8-bit programmable PWM
- ◆ 27bit programmable watch-dog timer (WDT)
- ◆ Full-Duplex UART
 - ✓ Automatic address recognition
- ◆ 11 interrupt sources (including 6 external sources)
 - ✓ Timer0/1/2, UART, WDT, 6 External
 - ✓ Four-level interrupt priority
- ◆ Wake-up from power-down mode by RESET or external interrupt 0/1 and watch-dog timer interrupt

Block diagram

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- ◆ Mobile Phone App.
 - ✓ Battery charger, Auto focusing controller (AFC)
- ◆ Security App.
 - ✓ Door lock, RFID card reader, Video phone
- ◆ Home App
 - ✓ Air cleaner, Microwave oven, Water purifier
- ◆ Automotive
 - ✓ Hands free, Radar detector
- ◆ Toy
 - ✓ Robot toy, line tracer

Performance characteristics

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	0.5 μm process	0.18 μm process
Frequency (MHz)	50 MHz (Max.)	100 MHz (Max.)
Gate count	20200 (Core) / 9900 (CPU)	
Active current	0.5 mA@5V per MHz	
Stop current	1 μA (Max.)	